PRODUCT REQUIREMENTS DOCUMENT (PRD)

WhatsApp Broadcast SaaS — "BroadcastHQ"

Version: 2.0 | **Date:** May 2026 | **Status:** Full Specification

EXECUTIVE SUMMARY

BroadcastHQ is a multi-tenant SaaS platform that enables small and medium-sized businesses to send highly personalised WhatsApp broadcast messages to their clients at scale. The platform differentiates itself from competitors such as WATI, AiSensy, and Respond.io through an AI-powered Excel/CSV import engine that intelligently identifies relevant columns without manual configuration, deep per-recipient message personalisation, a clean no-code campaign builder, and transparent per-message pricing with no hidden markups. The primary audience is non-technical small business owners who struggle with WhatsApp Business's native 256-contact broadcast limit and need an easy, affordable, and compliant path to reach thousands of clients.

1. PRODUCT OVERVIEW

1.1 Product Name

BroadcastHQ (working title — also suitable: PingBlast, ChatReach, WaveSend)

1.2 Vision Statement

Empower every small business to send WhatsApp messages that feel like they were written one-by-one — at the scale of thousands — without needing a developer or a marketing team.

1.3 Problem Statement

Small businesses face three acute pain points with WhatsApp outreach:

- **Scale barrier**: WhatsApp Business app limits broadcasts to 256 contacts; reaching 1,000–50,000 clients requires an API-backed platform.
- **Technical barrier**: Existing platforms (WATI, AiSensy, Respond.io) require API setup, template pre-approval, and developer knowledge that overwhelms non-technical users.
- **Personalisation gap**: Bulk tools send generic messages. Clients who receive "Dear Customer" messages ignore them. Recipients respond to messages that name them and reference their specific context.
- **Data friction**: Business owners store their client lists in Excel spreadsheets. Existing tools require clean CSV with specific column names, or manual CRM entry. There is no intelligent importer.

1.4 Core Value Proposition

BroadcastHQ solves all three problems:

- **Smart Excel Import**: Upload any Excel or CSV file. The AI engine analyses headers, data types, and patterns to automatically identify Name, Phone Number, Email, and any custom personalisation fields — no column mapping required.
- **Deep Personalisation Engine**: Every message sent is dynamically composed per recipient using their data — name, purchase history, custom attributes — so each message reads as if personally written.
- **No-Code Campaign Builder**: Create, preview, and send campaigns in under 5 minutes without any API knowledge.
- **Transparent Pricing**: Flat monthly SaaS fee + pass-through WhatsApp message cost at Meta's published rates. No markup on messages. No surprise bills.

1.5 Competitive Landscape

Platform	Monthly Price	Smart Import	Deep Personalisation	No-Code Setup	Message Markup
BroadcastHQ	£29–£149	■ AI-powered	■ Per-field dynamic	■ Wizard-based	■ None (pass-through)
WATI	$49–$299	■ Fixed CSV	■■ Basic variables	■ Moderate	■ Added
AiSensy	■999–■3,999/mo	■ Fixed CSV	■■ Basic variables	■ Good	■ Added
Respond.io	$79–$299	■ Manual CRM	■ Good	■■ Complex	■ None
Brevo	Free–custom	■ Email focus	■■ Basic	■ Good	■ Added
SendWo	Free–custom	■ Fixed CSV	■ None	■ Simple	■ None

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2. TARGET USERS

2.1 Primary Persona — "Sarah, the Salon Owner"

- **Age:** 28–45 | **Tech level:** Low-moderate
- **Business:** Beauty salon, local retailer, tutoring centre, estate agent, gym
- **Client list:** 200–5,000 clients stored in Excel (.xlsx) files
- **Current workflow:** Manually typing WhatsApp messages or using the 256-contact broadcast list
- **Pain points:** Spends hours messaging clients; messages are not personalised; no visibility on who read the message; cannot schedule campaigns
- **Goal:** Send appointment reminders, promotional offers, and seasonal greetings to all clients in 10 minutes

2.2 Secondary Persona — "James, the Agency Manager"

- **Age:** 25–40 | **Tech level:** Moderate-high
- **Business:** Digital marketing agency managing 10–30 small business clients
- **Goal:** Operate a white-label WhatsApp broadcast service for all clients from a single dashboard
- **Key need:** Multi-tenant workspace management, per-client analytics, role-based access

2.3 Tertiary Persona — "Dev Team / Enterprise"

- **Business:** Medium-sized business with existing CRM or custom data pipeline
- **Goal:** Use BroadcastHQ API to send campaigns programmatically
- **Key need:** REST API access, webhook integrations, advanced analytics export

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3. SCOPE & FEATURE ROADMAP

3.1 Phase 1 — MVP (Months 1–3)

Core features to achieve product-market fit:

- Multi-tenant SaaS architecture with tenant isolation
- AI-powered Excel/CSV intelligent import
- Contact management with deduplication
- WhatsApp template campaign creation and sending
- Bulk message queue (BullMQ + Redis)
- Delivery/read status webhooks and basic analytics dashboard
- Billing integration (Stripe) with three subscription tiers
- GDPR-compliant opt-out / unsubscribe management

3.2 Phase 2 — Growth (Months 4–6)

Features to increase retention and expand TAM:

- Deep personalisation engine with dynamic field variables
- Message scheduling (date/time picker, timezone-aware)
- Audience segmentation (tag-based, custom filter groups)
- Campaign A/B testing (two template variants, split audience)
- Team member invitations and role-based access (Owner, Admin, Editor, Viewer)
- Webhook outbound integration (Zapier, Make.com)

3.3 Phase 3 — Scale (Months 7–12)

Features to compete at enterprise tier:

- AI message writer (generate personalised message copy from a prompt + data)
- Two-way inbox (reply to broadcast respondents in a shared team inbox)
- Drip / sequence campaigns (multi-step message flows with time delays)
- WhatsApp Flows (interactive forms, product catalogues)
- CRM integrations (HubSpot, Pipedrive, Salesforce via Zapier)
- White-label agency dashboard
- Public REST API with developer documentation

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4. SYSTEM ARCHITECTURE

4.1 High-Level Architecture Diagram

```
■                      CLIENT BROWSER                          ■
■              React SPA (Next.js 14 App Router)               ■

                  ■ HTTPS / REST + WebSocket

■                   API GATEWAY (Cloudflare)                   ■
■        Rate limiting · DDoS protection · JWT validation      ■

■  Auth        ■   ■  Core API Server    ■   ■  Webhook Receiver   ■
■  Service     ■   ■  (Node.js/Fastify)  ■   ■  (Node.js/Fastify)  ■
■  (Supabase   ■   ■  Tenant-scoped      ■   ■  WhatsApp callbacks ■
■   Auth)      ■   ■  REST endpoints     ■   ■  Status updates     ■

■                    Supabase (PostgreSQL)                     ■
■      Row Level Security · UUID primary keys · pgcrypto      ■

■                 Message Queue (BullMQ + Redis)               ■
■   Campaign Worker · Retry Logic · Rate Limiter · Dead Letter Queue ■

■             WhatsApp Business API (Meta Cloud API)           ■
■     Template messages · Media · Status webhooks · Opt-out    ■
```

4.2 Technology Stack

Layer	Technology	Rationale
Frontend	Next.js 14 (App Router), TypeScript, Tailwind CSS v4	SSR for dashboard performance, strong typing
UI Components	shadcn/ui + Radix UI primitives	Accessible, customisable, production-grade
State Management	Zustand + React Query (TanStack)	Lightweight global state + server data caching
Backend API	Node.js + Fastify	High throughput, low latency, TypeScript native
Authentication	Supabase Auth (JWT + magic link + OAuth)	Managed auth, RLS integration
Database	Supabase (PostgreSQL 15)	Row Level Security, real-time subscriptions
File Storage	Supabase Storage + S3-compatible	Excel/CSV upload, media files
Message Queue	BullMQ + Redis (Upstash)	Reliable job processing, retries, rate limiting

Layer	Technology	Rationale
WhatsApp API	Meta Cloud API (direct)	No BSP markup; official, compliant
AI Engine	OpenAI GPT-4o mini (column detection + message writing)	Cost-effective, accurate for structured data tasks
Payments	Stripe (subscriptions + usage metering)	Industry standard; handles SCA/PSD2 for UK
Email	Resend (transactional)	Developer-friendly, reliable delivery
Monitoring	Sentry (errors) + PostHog (analytics) + Uptime Robot	Full observability
Deployment	Vercel (frontend) + Railway (backend + Redis)	Zero-downtime deploys, auto-scaling

4.3 Multi-Tenancy Design

Every database table includes a `tenant_id` UUID foreign key. Supabase Row Level Security policies enforce that all queries automatically filter to the authenticated user's tenant. No cross-tenant data leakage is architecturally possible without bypassing RLS. The API server validates JWT claims and injects `tenant_id` into every query — no client-supplied tenant IDs are trusted.

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5. DATABASE SCHEMA (FULL)

5.1 tenants

```
CREATE TABLE tenants (
  id               UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  name             TEXT NOT NULL,
  slug             TEXT UNIQUE NOT NULL,          -- used in URL e.g. /dashboard/my-salon
  plan             TEXT NOT NULL DEFAULT 'starter', -- starter | growth | pro | enterprise
  stripe_customer_id TEXT,
  whatsapp_phone_id  TEXT,                        -- Meta Cloud API phone ID
  whatsapp_access_token TEXT,                     -- encrypted at rest
  whatsapp_waba_id    TEXT,                        -- WhatsApp Business Account ID
  is_active          BOOLEAN NOT NULL DEFAULT true,
  created_at         TIMESTAMPTZ DEFAULT now(),
  updated_at         TIMESTAMPTZ DEFAULT now()
);
```

5.2 users

```
CREATE TABLE users (
  id               UUID PRIMARY KEY REFERENCES auth.users(id),
  tenant_id        UUID NOT NULL REFERENCES tenants(id) ON DELETE CASCADE,
  email            TEXT NOT NULL,
  full_name        TEXT,
  role             TEXT NOT NULL DEFAULT 'editor', -- owner | admin | editor | viewer
  avatar_url       TEXT,
  last_login_at    TIMESTAMPTZ,
  created_at       TIMESTAMPTZ DEFAULT now()
);
```

5.3 contacts

```
CREATE TABLE contacts (
  id              UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  tenant_id       UUID NOT NULL REFERENCES tenants(id) ON DELETE CASCADE,
  phone_number    TEXT NOT NULL,                  -- E.164 format e.g. +447911123456
  full_name       TEXT,
  email           TEXT,
  custom_fields   JSONB DEFAULT '{}',             -- flexible key-value store for imported columns
  tags            TEXT[] DEFAULT '{}',
  opted_out       BOOLEAN NOT NULL DEFAULT false,
  opted_out_at    TIMESTAMPTZ,
  created_at      TIMESTAMPTZ DEFAULT now(),
  updated_at      TIMESTAMPTZ DEFAULT now(),
  UNIQUE(tenant_id, phone_number)
);

CREATE INDEX idx_contacts_tenant ON contacts(tenant_id);
CREATE INDEX idx_contacts_tags ON contacts USING GIN(tags);
CREATE INDEX idx_contacts_custom_fields ON contacts USING GIN(custom_fields);
```

5.4 contact_lists

```
CREATE TABLE contact_lists (
  id              UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  tenant_id       UUID NOT NULL REFERENCES tenants(id) ON DELETE CASCADE,
  name            TEXT NOT NULL,
  description     TEXT,
  contact_count   INTEGER DEFAULT 0,
  created_at      TIMESTAMPTZ DEFAULT now()
);

CREATE TABLE contact_list_members (
  list_id         UUID NOT NULL REFERENCES contact_lists(id) ON DELETE CASCADE,
  contact_id      UUID NOT NULL REFERENCES contacts(id) ON DELETE CASCADE,
  PRIMARY KEY(list_id, contact_id)
);
```

5.5 campaigns

```
CREATE TABLE campaigns (
  id                  UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  tenant_id           UUID NOT NULL REFERENCES tenants(id) ON DELETE CASCADE,
  name                TEXT NOT NULL,
  status              TEXT NOT NULL DEFAULT 'draft', -- draft | scheduled | running | paused | completed | fa
  template_id         TEXT,                          -- Meta template name
  template_language   TEXT DEFAULT 'en',
  message_body        TEXT,                          -- personalisation template string
  personalisation_map JSONB DEFAULT '{}',            -- field_var -> contact_field mapping
  audience_list_id    UUID REFERENCES contact_lists(id),
  audience_filters    JSONB DEFAULT '{}',            -- segment filter conditions
  scheduled_at        TIMESTAMPTZ,
  started_at          TIMESTAMPTZ,
  completed_at        TIMESTAMPTZ,
  total_recipients    INTEGER DEFAULT 0,
  sent_count          INTEGER DEFAULT 0,
  delivered_count     INTEGER DEFAULT 0,
  read_count          INTEGER DEFAULT 0,
  failed_count        INTEGER DEFAULT 0,
  created_by          UUID REFERENCES users(id),
  created_at          TIMESTAMPTZ DEFAULT now(),
```

```
  updated_at          TIMESTAMPTZ DEFAULT now()
);

CREATE INDEX idx_campaigns_tenant ON campaigns(tenant_id);
CREATE INDEX idx_campaigns_status ON campaigns(status);
```

5.6 messages

```
CREATE TABLE messages (
  id                  UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  tenant_id           UUID NOT NULL REFERENCES tenants(id) ON DELETE CASCADE,
  campaign_id         UUID NOT NULL REFERENCES campaigns(id) ON DELETE CASCADE,
  contact_id          UUID NOT NULL REFERENCES contacts(id),
  phone_number        TEXT NOT NULL,
  message_body        TEXT,                          -- rendered, personalised message
  wa_message_id       TEXT,                          -- WhatsApp message ID from Meta API
  status              TEXT NOT NULL DEFAULT 'queued', -- queued | sent | delivered | read | failed | opted_ou
  error_code          TEXT,
  error_message       TEXT,
  sent_at             TIMESTAMPTZ,
  delivered_at        TIMESTAMPTZ,
  read_at             TIMESTAMPTZ,
  failed_at           TIMESTAMPTZ,
  created_at          TIMESTAMPTZ DEFAULT now()
);

CREATE INDEX idx_messages_campaign ON messages(campaign_id);
CREATE INDEX idx_messages_wa_id ON messages(wa_message_id);
CREATE INDEX idx_messages_status ON messages(status);
```

5.7 imports

```
CREATE TABLE imports (
  id              UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  tenant_id       UUID NOT NULL REFERENCES tenants(id) ON DELETE CASCADE,
  file_name       TEXT NOT NULL,
  file_url        TEXT NOT NULL,                    -- Supabase Storage URL
  file_size_bytes INTEGER,
  row_count       INTEGER,
  status          TEXT NOT NULL DEFAULT 'pending', -- pending | processing | completed | failed
  ai_column_map   JSONB DEFAULT '{}',               -- AI-detected column mappings
  contacts_created INTEGER DEFAULT 0,
  contacts_updated INTEGER DEFAULT 0,
  contacts_skipped INTEGER DEFAULT 0,
  error_log       JSONB DEFAULT '[]',
  created_by      UUID REFERENCES users(id),
  created_at      TIMESTAMPTZ DEFAULT now(),
  completed_at    TIMESTAMPTZ
);
```

5.8 billing_events

```
CREATE TABLE billing_events (
  id              UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  tenant_id       UUID NOT NULL REFERENCES tenants(id) ON DELETE CASCADE,
  event_type      TEXT NOT NULL,                    -- subscription_created | invoice_paid | message_charge | e
  stripe_event_id TEXT,
  amount_pence    INTEGER,                          -- in GBP pence
  currency        TEXT DEFAULT 'gbp',
```

```
  metadata        JSONB DEFAULT '{}',
  created_at      TIMESTAMPTZ DEFAULT now()
);
```

6. API SPECIFICATION

6.1 Authentication

All API requests require a Bearer JWT token issued by Supabase Auth. Tokens are short-lived (1 hour) and refreshed automatically by the frontend client. Service-to-service calls use a server-side service role key stored as an environment secret.

```
Authorization: Bearer <jwt_token>
X-Tenant-ID: <tenant_uuid>   (validated against JWT claims)
```

6.2 Contacts API

Method	Endpoint	Description
GET	/api/v1/contacts	List contacts (paginated, filterable)
POST	/api/v1/contacts	Create a single contact
GET	/api/v1/contacts/:id	Get contact by ID
PATCH	/api/v1/contacts/:id	Update contact fields
DELETE	/api/v1/contacts/:id	Soft-delete contact
POST	/api/v1/contacts/bulk	Bulk upsert contacts
POST	/api/v1/contacts/import	Upload Excel/CSV file for AI import
GET	/api/v1/contacts/import/:id	Poll import job status

6.3 Campaigns API

Method	Endpoint	Description
GET	/api/v1/campaigns	List all campaigns
POST	/api/v1/campaigns	Create new campaign
GET	/api/v1/campaigns/:id	Get campaign details + stats
PATCH	/api/v1/campaigns/:id	Update campaign (draft only)
DELETE	/api/v1/campaigns/:id	Delete campaign (draft only)
POST	/api/v1/campaigns/:id/send	Start sending campaign immediately
POST	/api/v1/campaigns/:id/schedule	Schedule campaign for future time
POST	/api/v1/campaigns/:id/pause	Pause running campaign
POST	/api/v1/campaigns/:id/resume	Resume paused campaign
GET	/api/v1/campaigns/:id/messages	List messages for campaign

6.4 Analytics API

Method	Endpoint	Description
GET	/api/v1/analytics/overview	Dashboard summary stats
GET	/api/v1/analytics/campaigns/:id	Detailed campaign analytics
GET	/api/v1/analytics/contacts	Contact growth over time
GET	/api/v1/analytics/delivery	Delivery rate trends

6.5 Webhook Endpoints (Inbound from Meta)

Method	Endpoint	Description
GET	/webhook/whatsapp	Meta verification challenge
POST	/webhook/whatsapp	Status updates (sent/delivered/read/failed) + opt-outs

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7. AI-POWERED EXCEL IMPORT ENGINE

7.1 Overview

The import engine is the primary differentiator. Business owners upload any Excel or CSV file — with no pre-defined column naming convention — and the system automatically detects which columns represent phone numbers, names, emails, and any other personalisation-relevant data.

7.2 Detection Algorithm

Step 1 — File Parsing

- Accept `.xlsx`, `.xls`, `.csv` files up to 25 MB
- Use `xlsx` (SheetJS) for Excel parsing; `csv-parse` for CSV
- Auto-detect header row (first non-empty row with mostly text values)
- Extract first 50 rows as sample data for AI analysis

Step 2 — AI Column Classification

Send a structured prompt to GPT-4o mini with:

- Column headers
- Sample values from first 50 rows per column
- Instruction to return a JSON mapping

```
{
  "phone": "column_index_or_null",
  "name": "column_index_or_null",
  "first_name": "column_index_or_null",
  "last_name": "column_index_or_null",
  "email": "column_index_or_null",
  "custom_fields": [
```

```
    { "column_index": 4, "suggested_key": "appointment_date", "data_type": "date" },
    { "column_index": 5, "suggested_key": "service_type", "data_type": "string" }
  ]
}
```

Step 3 — Phone Number Normalisation

- Strip all non-digit characters
- Apply `libphonenumber-js` to validate and convert to E.164 format
- Use tenant's configured default country code for local numbers (e.g., `+44` for UK)
- Flag invalid numbers in the import error log

Step 4 — Deduplication

- Match contacts by normalised phone number within the tenant's namespace
- Existing contacts: update `custom_fields` with new columns; preserve opt-out status
- New contacts: insert with all detected fields

Step 5 — User Review & Confirm

- Present the AI-detected mapping to the user in a review UI
- Allow manual override of any column assignment
- Show preview of how first 5 contacts will look post-import
- User confirms → background job processes full file

7.3 Supported Import File Formats

Format	Max Size	Max Rows	Notes
.xlsx (Excel 2007+)	25 MB	100,000	Multi-sheet: first sheet used
.xls (Excel 97–2003)	10 MB	50,000	Converted via SheetJS
.csv (UTF-8)	25 MB	150,000	Auto-detect delimiter (comma, semicolon, tab)
.csv (with BOM)	25 MB	150,000	BOM stripped automatically

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8. PERSONALISATION ENGINE

8.1 Template Variable Syntax

Messages use double-curly-brace syntax for dynamic fields:

```
Hi {{first_name}}, your appointment at {{business_name}} is confirmed for {{appointment_date}} at {{appointme
Reply STOP to unsubscribe.
```

8.2 Available Variable Sources

Variable Type	Source	Example
Contact fields	contacts.custom_fields	`{{first_name}}`, `{{appointment_date}}`

Variable Type	Source	Example
Tenant fields	tenants table	`{{business_name}}`, `{{business_phone}}`
System fields	Generated at send time	`{{opt_out_link}}`, `{{today_date}}`
Campaign fields	campaigns table	`{{campaign_name}}`

8.3 Fallback Values

Each variable supports a fallback value for when the contact field is empty:

```
Hi {{first_name | "there"}}, ...
```

If `first_name` is empty, renders as: `Hi there, ...`

8.4 WhatsApp Template Compliance

Meta requires pre-approved templates for outbound marketing messages. BroadcastHQ manages this by:

- User creates message body in the campaign builder
- System automatically generates a Meta-compliant template payload with `{{1}}`, `{{2}}` parameter format
- Template is submitted to Meta for approval (typically 1–24 hours)
- Campaign is unlocked for sending once template status is `APPROVED`

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9. BILLING & PRICING MODEL

9.1 Subscription Tiers

Plan	Price/Month	Contacts	Campaigns/Month	Team Members	Key Features
Starter	£29	Up to 2,000	10	1	AI import, basic analytics
Growth	£79	Up to 10,000	Unlimited	5	Scheduling, segmentation, A/B testing
Pro	£149	Up to 50,000	Unlimited	15	API access, white-label, priority support
Enterprise	Custom	Unlimited	Unlimited	Unlimited	SLA, dedicated support, custom integrations

9.2 WhatsApp Message Costs

BroadcastHQ passes through Meta's WhatsApp Business API message costs at **zero markup**:

- **Marketing messages (UK):** ~£0.048 per conversation (24-hour window)
- **Utility messages (UK):** ~£0.0085 per conversation
- **Authentication messages:** ~£0.0187 per conversation

A "conversation" = all messages to the same recipient within a 24-hour window. The platform displays the estimated cost before each campaign is sent.

9.3 Billing Implementation (Stripe)

- Subscription fees: Stripe Subscriptions with monthly billing cycle
- WhatsApp message costs: Stripe Metered Billing (usage-based) — recorded as Stripe usage records at end of month
- Payment methods: Card (Visa, Mastercard, Amex), SEPA Direct Debit, BACS Direct Debit (UK)
- Invoices: Auto-generated by Stripe, emailed to billing contact
- Failed payments: Retry logic → 3 attempts → account downgrade to free tier

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10. SECURITY & COMPLIANCE

10.1 Data Security

- **Encryption at rest:** All database data encrypted via Supabase/PostgreSQL AES-256
- **Encryption in transit:** TLS 1.3 enforced on all endpoints
- **WhatsApp credentials:** Encrypted using `pgcrypto` before storage; decrypted only in memory during API calls
- **API key management:** Hashed (bcrypt) before storage; shown once on creation
- **Secrets management:** All API keys stored in Railway/Vercel environment variables; never in codebase

10.2 GDPR Compliance

- **Lawful basis:** Consent or Legitimate Interest — business owner is responsible for their contact list
- **Opt-out management:** Every message must contain an opt-out mechanism (STOP keyword or unsubscribe link). Opted-out contacts are permanently blocked from future sends.
- **Data residency:** EU/UK data residency option on Pro/Enterprise plans (Supabase EU region)
- **Right to erasure:** Contact deletion endpoint permanently removes all personal data; audit log retained (no PII)
- **Data retention:** Message logs retained 90 days (Starter), 1 year (Growth/Pro), configurable (Enterprise)
- **DPA (Data Processing Agreement):** Available for all paid plans

10.3 WhatsApp Business Policy Compliance

- All campaigns require WhatsApp Business API approval
- Only businesses with verified Meta Business Manager accounts may connect
- Platform enforces Meta's messaging policies: no spam, no unsolicited messages, mandatory opt-out
- Abuse detection: automated flagging of tenants with >5% block rate or unusual send volumes

10.4 Rate Limiting

- API: 100 requests/min per tenant (Starter), 500/min (Growth), 2,000/min (Pro)
- WhatsApp sends: 80 messages/sec per phone number (Meta's default limit)

- Campaign queue automatically rate-limits outbound sends to stay within Meta's per-second limits
- Tenant-level daily send cap enforced based on plan

11. USER EXPERIENCE (UX) FLOWS

11.1 Onboarding Flow

```
1. Sign up with email / Google OAuth
2. Create workspace (business name, type, country)
3. Connect WhatsApp Business Account (guided Meta OAuth flow)
4. Import first contact list (drag-and-drop Excel upload)
5. AI reviews and maps columns → user confirms
6. Create first campaign (template selector → message editor → audience picker)
7. Preview → Send or Schedule
8. View live delivery stats on dashboard
```

11.2 Campaign Creation Flow

```
1. New Campaign → Name the campaign
2. Select / create WhatsApp template
3. Write personalised message body with {{variable}} tags
4. Preview rendered message for 3 sample contacts
5. Select audience: All Contacts | Specific List | Segment Filter
6. Estimated cost preview (contact count × message rate)
7. Send Now OR Schedule (date/time picker)
8. Confirm → Campaign enters queue
9. Live progress bar with sent/delivered/read counters
```

11.3 Import Flow (Detailed)

```
1. Contacts → Import → Drag & drop or browse file
2. Upload progress bar (chunked upload for large files)
3. AI processing spinner ("Analysing your file...")
4. Review screen: AI-detected column mappings shown as cards
   - Phone Number: Column B (Mobile) ■
   - First Name: Column A (Name) ■
   - Custom: Column D → appointment_date ■
5. User can drag/reassign any mapping
6. Preview: 5 contacts shown as they will appear
7. Confirm Import → background job runs
8. Email notification when complete + import summary
```

12. NOTIFICATIONS & ALERTS

12.1 In-App Notifications

- Campaign completed (with summary stats)
- Import completed / import failed

- WhatsApp template approved / rejected by Meta
- Team member invitation accepted
- Monthly usage summary

12.2 Email Notifications (via Resend)

- Welcome email on signup
- Campaign completion report
- Import completion summary
- Billing invoices and payment receipts
- Account suspension warnings
- Password reset / magic link

12.3 Webhook Outbound (Phase 2+)

Tenants on Growth+ can configure outbound webhooks to be called on:

- `campaign.completed`
- `contact.opted_out`
- `message.delivered`
- `message.read`
- `message.failed`

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13. ANALYTICS & REPORTING

13.1 Dashboard Metrics

- Total messages sent (this month / all time)
- Delivery rate (%) — industry benchmark: >95%
- Read rate (%) — industry benchmark: 60–80%
- Opt-out rate (%) — alert if >2%
- Active campaigns
- Contact list growth chart (30-day rolling)

13.2 Per-Campaign Analytics

- Funnel: Sent → Delivered → Read → Replied
- Delivery timeline chart (messages sent per minute)
- Failed messages with error reason breakdown
- Geographic distribution (if available from Meta)
- A/B test winner (Phase 2) — which variant achieved higher read rate

13.3 Export

- Export campaign analytics as CSV

- Export contact list (with custom fields) as Excel
- Export message logs (filtered by date/status) as CSV

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14. INTEGRATION SPECIFICATIONS

14.1 Meta WhatsApp Cloud API Integration

Connection Setup:

- User clicks "Connect WhatsApp" in settings
- OAuth redirect to Meta Business Manager
- User selects WhatsApp Business Account and phone number
- BroadcastHQ stores `phone_number_id` and `access_token` (encrypted)
- Webhook registered: `POST /webhook/whatsapp` receives all status callbacks

Template Management:

- Templates are fetched from Meta API on login and cached in Redis (15 min TTL)
- New templates submitted via `POST /v18.0/{waba_id}/message_templates`
- Approval status polled every 5 minutes for pending templates

Message Sending:

```
POST https://graph.facebook.com/v18.0/{phone_number_id}/messages
{
  "messaging_product": "whatsapp",
  "to": "+447911123456",
  "type": "template",
  "template": {
    "name": "appointment_reminder",
    "language": { "code": "en" },
    "components": [{
      "type": "body",
      "parameters": [
        { "type": "text", "text": "Sarah" },
        { "type": "text", "text": "2026-05-15 at 14:00" }
      ]
    }]
  }
}
```

14.2 Stripe Integration

- **Checkout:** Stripe Checkout hosted page for plan upgrades
- **Webhooks:** `invoice.paid`, `invoice.payment_failed`, `customer.subscription.deleted`
- **Metered billing:** Usage reported via `POST /v1/subscription_items/{si_id}/usage_records`
- **Customer portal:** Stripe Customer Portal for self-serve billing management

14.3 Zapier / Make.com Integration (Phase 2)

- BroadcastHQ listed as a Zapier app with triggers and actions
- Triggers: `New Contact`, `Campaign Completed`, `Contact Opted Out`

- Actions: `Add Contact`, `Add to List`, `Send Message`

---

15. PERFORMANCE REQUIREMENTS

Metric	Target	Notes
API response time (p95)	< 300ms	Excluding WhatsApp API call time
Dashboard load time	< 2 seconds	First Contentful Paint
Campaign send throughput	80 messages/sec	Per phone number (Meta limit)
File import processing	< 60 seconds	For 10,000-row Excel file
Uptime SLA	99.9%	Excluding planned maintenance
Webhook processing	< 5 seconds	From Meta callback to DB update
Queue capacity	5,000,000 jobs	Before Redis memory pressure

---

16. LAUNCH PLAN

16.1 Pre-Launch (Months 1–2)

- [ ] Core MVP development (see Section 3.1)
- [ ] Meta Business Verification (required for WhatsApp API access)
- [ ] Stripe account setup and UK KYC verification
- [ ] Beta waitlist landing page live at broadcasthq.com
- [ ] Recruit 20 beta testers from local UK small business networks

16.2 Beta Launch (Month 3)

- [ ] Invite beta testers with free 3-month Growth plan access
- [ ] Gather feedback via in-app survey (PostHog) + weekly calls
- [ ] Prioritise bug fixes and UX improvements
- [ ] Finalise pricing based on beta user willingness-to-pay data

16.3 Public Launch (Month 4)

- [ ] ProductHunt launch (Tuesday 00:01 PST for maximum votes)
- [ ] Outreach to UK small business Facebook groups and WhatsApp communities
- [ ] Partnerships with local business accountants and marketing agencies
- [ ] Google Ads campaign (target: "whatsapp bulk message uk", "whatsapp business api uk")
- [ ] Blog content: "How to send WhatsApp messages to all your clients" (SEO)

16.4 Growth Milestones

Milestone	Target Date	KPI
MVP live	Month 3	0 → 1 paying customer
Beta closed	Month 4	20 active beta users, NPS > 40
£1k MRR	Month 5	~30 Starter / 10 Growth customers
£5k MRR	Month 8	~100 customers
£10k MRR	Month 12	~200 customers
Series A ready	Month 18	£25k MRR, <5% monthly churn

17. RISKS & MITIGATIONS

Risk	Likelihood	Impact	Mitigation
Meta rejects WhatsApp API application	Medium	Critical	Apply via official BSP partner route; keep alternative BSP (360dialog) as backup
Meta changes API pricing	Medium	High	Monitor Meta developer blog; pricing passed through to customers removes margin risk
GDPR enforcement action	Low	High	Ensure DPA with all tenants; clear consent documentation; appoint UK DPO
Competitor copies AI import feature	High	Medium	Build moat through integrations, brand, and switching costs
Redis/Supabase outage during send	Low	High	Multi-region Redis (Upstash); database replication enabled; retry queue for failed sends
Stripe account closure	Very Low	Critical	Keep PayPal Braintree as secondary payment processor
Spam abuse by tenant	Medium	High	Automated abuse detection; block rate monitoring; manual review threshold

18. SUCCESS METRICS (KPIs)

18.1 Product KPIs

- **Monthly Active Tenants (MAT):** Number of tenants who sent at least 1 campaign in the month
- **Campaign Send Rate:** % of registered tenants who sent a campaign this month
- **Messages Sent:** Total messages dispatched across all tenants
- **Delivery Rate:** % of sent messages with `delivered` or `read` status

- **Import Success Rate:** % of imports completing without user-reported errors
- **AI Column Detection Accuracy:** % of columns correctly auto-detected (target: >90%)

18.2 Business KPIs

- **MRR (Monthly Recurring Revenue):** Target £10k by Month 12
- **Churn Rate:** Target <5% monthly
- **CAC (Customer Acquisition Cost):** Target <£50 per customer
- **LTV (Lifetime Value):** Target >£500 per customer
- **NPS (Net Promoter Score):** Target >50

APPENDIX A — Environment Variables

```
# Supabase
SUPABASE_URL=
SUPABASE_ANON_KEY=
SUPABASE_SERVICE_ROLE_KEY=

# WhatsApp / Meta
META_APP_ID=
META_APP_SECRET=
META_WEBHOOK_VERIFY_TOKEN=

# OpenAI (AI Import Engine)
OPENAI_API_KEY=

# Stripe
STRIPE_SECRET_KEY=
STRIPE_PUBLISHABLE_KEY=
STRIPE_WEBHOOK_SECRET=

# Redis (Upstash)
REDIS_URL=
REDIS_TOKEN=

# Resend (Email)
RESEND_API_KEY=

# App
NEXT_PUBLIC_APP_URL=
API_BASE_URL=
JWT_SECRET=
ENCRYPTION_KEY=   # 32-byte key for WhatsApp token encryption
```

APPENDIX B — Glossary

Term	Definition
BSP	Business Solution Provider — a Meta-authorised WhatsApp API reseller

Term	Definition
WABA	WhatsApp Business Account — the Meta account that owns a WhatsApp phone number
Template	A pre-approved message structure required for outbound (non-session) WhatsApp messages
Conversation Window	A 24-hour billing period opened when a message is sent or received
E.164	International phone number format: +[country code][number], e.g. +447911123456
Tenant	A single business workspace within the BroadcastHQ multi-tenant platform
RLS	Row Level Security — PostgreSQL feature enforcing per-row access policies
MRR	Monthly Recurring Revenue
LTV	Lifetime Value of a customer
CAC	Customer Acquisition Cost

Document prepared by: BroadcastHQ Product Team

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